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Stability and bifurcation analysis of Generation IV reactors via point reactor models with temperature reactivity feedback

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ABSTRACT

While the stability of present-day reactors is well studied, stability analysis of Generation IV reactors is in its very beginning. In this paper, the stability of point nuclear reactor models with temperature reactivity feedbacks is analyzed. The examined models are referred to as G-L reactor models, which are point reactor models with G delayed neutron-emission groups, L temperature lumps (regions) and a linear reactivity feedback for each temperature. G-L reactor models are applicable for modeling four out of the six Generation IV reactor types (Gas Fast Reactor, Lead Fast Reactor, Sodium Fast Reactor and Very High Temperature Reactor).

New analytical stability and bifurcation results of general G-L reactor models and of the G1-L2 reactor model are derived. These analytical results add a different perspective to the previous studies of Generation IV reactors, most of which are based only on numerical analysis. In particular, the codimension-1 bifurcations of general G-L reactor models are characterized, and a classification of the linear stability region in the G1-L2 model is derived.

Then, stability and bifurcation analysis of a G1-L2 model of a Generation IV HTR-type microreactor is performed, using both analytical methods and the numerical continuation method. The nominal design point of this microreactor is shown to be globally asymptotically stable, while in other regions of the parameter space, a bistability of the operating steady state and a limit cycle, or of two limit cycles, is demonstrated.

1. Introduction

Generation IV nuclear reactors are new advanced reactor designs that represent a considerable departure from Generation II and III nuclear reactors, with major leaps in sustainability, safety, economics, proliferation resistance and physical protection (GIF, 2018). Six Generation IV reactor types are being considered: Gas-cooled Fast Reactor (GFR), Lead-cooled Fast Reactor (LFR), Molten Salt Reactor (MSR), Super-Critical Water-cooled Reactor (SCWR), Sodium-cooled Fast Reactor (SFR), and Very High Temperature Reactor (VHTR). The types with the most research and development progress to date are SFR and VHTR (GIF, 2018).

A significant part of the studies on nuclear reactor stability in the past concentrated on two types of instabilities observed in Generation II reactors: Boiling Water Reactor (BWR) oscillations (due to coupling between thermal-hydraulics and neutronics) (NEA, 1997) and Xenon oscillations (due to coupling between neutronics and reactor burnup) (Bell and Glasstone, 1970). In contrast, the study of nuclear reactor stability of Generation IV reactors is currently in its beginning. As few Generation IV prototype reactors are (or were) in operation, the focus of the stability studies is theoretical. Stability of six reference Generation IV reactor designs, representing the six different Generation IV reactor types, was studied in Cervi et al. (2018). Other studies concentrated on a single Generation IV type, e.g. LFR (Bortot et al., 2013; Shahzad et al., 2018), MSR (Guerrieri et al., 2013; Singh et al., 2018), SCWR (Yi et al., 2004; Zhao et al., 2008) and SFR (Kim et al., 2016).

Previous stability studies of Generation IV reactors have mostly focused on a specific reactor type, and they usually used numerical methods. In this paper, a different approach is taken: stability of simplified reactor models, applicable to four out of the six Generation IV reactor types, is studied via both analytical and numerical methods. Analytical understanding of the stability of these simplified models can provide more insight into the stability of realistic Generation IV reactors. In the 1950s through the 1970s, many analytical studies of

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simplified reactor models were published (Akcasu et al., 1971; Hetrick, 1971). This paper builds upon these studies, and adds new and important analytical results.

The simplified reactor models studied in this paper are referred to here as "G-L reactor models". A G-L reactor model can be described shortly as a point reactor model with G delayed neutron-emission groups (delay groups), L temperature lumps (regions) and a linear temperature reactivity feedback. It is a simplified model, which couples neutronics and thermal-hydraulics, and thus can theoretically predict instabilities due to neutronics and thermal-hydraulics coupling. It is important to note that, since G-L reactor models are simplified, more detailed models are necessary to better predict instabilities.

G-L reactor models are applicable to reactor cores with a solid fuel and with fluids in which the fluid flow and heat transfer are approximately decoupled. Thus, G-L reactor models can model four out of the six Generation IV reactor types: GFR, LFR, SFR and VHTR. It cannot model MSRs, in which the fuel is a liquid, and it cannot model SCWR because the supercritical water admits a pseudo two-phase flow, with a strong coupling between fluid flow and heat transfer.

Stability of G-L reactor models have been studied in the past, mainly through analytical methods, e.g. G0-L2 (G-L reactor model with $G = 0$ and $L = 2$) (Vreeke and Sandquist, 1970), G0-L3 (Lellouche, 1966), G1-L2 (Miida and Suda, 1961; Schmidt and Hetrick, 1970) and general G-L reactor models (Ergen et al., 1957). In recent years, stability of G-L reactor models of Generation IV reactors was studied numerically: in Cervi et al. (2018) (for GFR, LFR, SFR and VHTR) and in Shahzad et al. (2018) (for LFR). This paper unifies these models under a single generalized framework, in which more general studies can be carried, and connections between the results of different models in the family (or the same model with different parameters) can be analyzed.

In Section 2, essential background for stability analysis of G-L reactor models is given. In Section 3, G-L reactor models in regular form are defined. In Sections 4 and 5, new analytical linear stability results are derived for general G-L models and specifically for the G1-L2 model, respectively. In Section 6, stability and bifurcation analysis of a G1-L2 reactor model of a Generation IV HTR-type microreactor is performed, using both analytical methods and the numerical continuation method.

2. Background

2.1. Point reactor model with linear reactivity feedback

G-L reactor models are a special case of the more general point reactor model with linear reactivity feedback (Akcasu et al., 1971; Hetrick, 1971):

$$\frac{d}{dt}P(t) = \frac{\rho(t)-\beta}{\Lambda}P(t) + \sum_{k=1}^{G}\lambda_k C_k(t) \ , \tag{1}$$

$$\frac{d}{dt}C_k(t) = \frac{\beta_k}{\Lambda}P(t) - \lambda_k C_k(t) \ , \tag{2}$$

$$\rho(t) = \rho_e(t) + \rho_{fb}[P](t) = \rho_e(t) + \int_{-\infty}^{t} h(t-t')P(t')dt' \ . \tag{3}$$

Eqs. (1) and (2) are the well-known point reactor equations for the reactor power $P(t)$ with G delay groups[1] (Bell and Glasstone, 1970). Eq. (3) is the reactivity equation, where ρ_e is the external reactivity at zero power, $\rho_{fb}[P]$ is a linear reactivity feedback functional of the reactor power, and $h(t)$ is the feedback impulse response function. The external reactivity at zero power, ρ_e, is a function of the state of the core, including for example the position of the control rods. The nonlinearity of this model stems only from the product $\rho(t)P(t)$ in Eq. (1).

If ρ_e is time-independent, then the point reactor model with linear reactivity feedback is an autonomous dynamical system (meaning that all of the parameters in the model are time-independent). In that case, there are at most two steady states: the shutdown steady state, and the operating steady state. The shutdown steady state (denoted by a 0 subscript) is given by $P_0 = C_{k,0} = 0$ and $\rho_0 = \rho_e$. The operating steady state (denoted by a 1 subscript) is given by $P_1 = -\frac{\rho_e}{H(0)}$, $C_{k,1} = \frac{\beta_k}{\Lambda\lambda_k}P_1$ and $\rho_1 = 0$, where $H(0) = \int_0^\infty h(t')dt'$. The operating steady state exists only if $\frac{\rho_e}{H(0)} < 0$.

It is usually assumed that the point reactor model with linear reactivity feedback is a finite-dimensional dynamical system, i.e. the reactivity equation, Eq. (3), can be written as follows:

$$\frac{d}{dt}x_i(t) = c_i P(t) + \sum_{j=1}^{N} A_{ij}x_j(t) \ , \tag{4}$$

$$\rho(t) = \rho_e(t) + \rho_{fb}(t) = \rho_e(t) + \sum_{i=1}^{N}\alpha_i x_i(t) \ , \tag{5}$$

where α_i is the reactivity coefficient of the dynamical variable $x_i(t)$, and Eq. (4) describes a general linear dynamics of these N variables.

2.2. Linear stability

The linear stability, also called asymptotic stability, of the steady states of the autonomous finite-dimensional model, can be analyzed using two equivalent approaches: dynamical systems theory (Guckenheimer and Holmes, 1983) and linear control theory (Golnaraghi and Kuo, 2010). In the approach of dynamical systems theory, the steady state is linearly stable if and only if the Jacobian of the autonomous dynamical system, evaluated at the steady state, has only eigenvalues with negative real value. In the linear control theory approach, the steady state is linearly stable if and only if the transfer function of the (non-autonomous) system, linearized around the steady state, has only poles with negative real value.

For the point reactor model with linear reactivity feedback, the shutdown steady state is linearly stable if and only if $\rho_e < 0$. The condition for the linear stability of the operating steady state is generally far more complicated. The linear stability of the operating steady state is commonly analyzed via linear control theory, by expressing the system (closed-loop) transfer function, $G(s)$, using the zero-power transfer function, $\bar{G}_0(s)$, and the feedback transfer function, $H(s)$, as follows (Bell and Glasstone, 1970):

$$\bar{G}(s) = \frac{1}{P_1}\frac{\delta P(s)}{\delta\rho_e(s)} = \frac{\bar{G}_0(s)}{1 - P_1\bar{G}_0(s)H(s)} \ , \tag{6}$$

$$\bar{G}_0(s) = \frac{1}{P_1}\frac{\delta P(s)}{\delta\rho(s)} = \frac{1}{s\left(\Lambda + \sum_{k=1}^{G}\frac{\beta_k}{s+\lambda_k}\right)} \ , \tag{7}$$

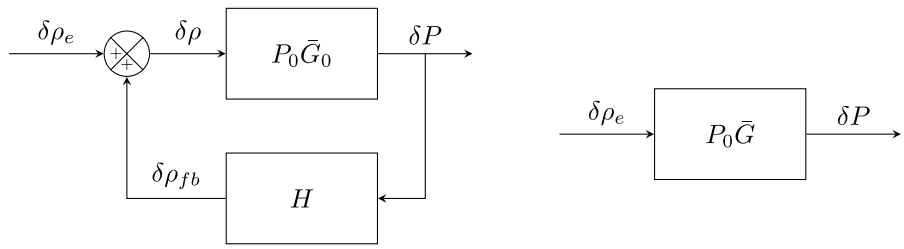
$$H(s) = \frac{\delta\rho_{fb}(s)}{\delta P(s)} \ , \tag{8}$$

which can be straightforwardly derived by Laplace transforming (to the s-domain) the model equations, linearized around the operating steady state with power P_1, where $\delta\rho_e(t)$ is the input variable and $\delta P(t)$ is the output variable (δ denotes a deviation of a variable from its steady state value). Note that the feedback transfer function $H(s)$ is the Laplace transform of the feedback impulse response function $h(t)$.

The system transfer function $G(s)$ is commonly described by a closed-loop block diagram or an equivalent simplified block diagram, as shown in Fig. 1. In this paper, the positive feedback convention is used: $\rho = \rho_e + \rho_{fb}$ (and not $\rho = \rho_e - \rho_{fb}$), which is not the standard convention in linear control theory, but is commonly used in reactor stability analysis (Bell and Glasstone, 1970).

The open-loop transfer function is given by $\bar{L}(s) = -P_1\bar{G}_0(s)H(s)$, and the characteristic equation is $1 + \bar{L}(s) = 0$. The roots of the

[1] Standard symbols are used: P for reactor power, C_k for effective delayed neutron precursors of delay group k, ρ for reactivity, β_k for effective delayed neutron of delay group k, $\beta = \sum_{k=1}^{G}\beta_k$ for total effective delayed neutron fraction, Λ for neutron generation time, and λ_k for decay constant of delay group k.



(a) Closed-Loop Block Diagram (b) Equivalent Block Diagram

Fig. 1. Equivalent Block Diagrams of system transfer function, $G(s)$, of the point reactor model with linear reactivity feedback.

characteristic equation are generically the poles of the system transfer function $\bar{G}(s)$, and thus linear stability of the operating steady state is equivalent to all the roots of the characteristic equation having negative real value.

If $H(s)$ only has poles with negative real value, then a necessary condition for the linear stability of the operating steady state, called the static stability condition, is given by $H(0) < 0$ (Bell and Glasstone, 1970). It is called static stability condition, because $H(0) = \frac{\Delta\rho_{fb}}{\Delta P}$ is the static power reactivity coefficient: it is the static change in feedback reactivity which accompanies a static change in power. It can be shown that if the static stability condition is satisfied, then the operating steady state is linearly stable at sufficiently low powers. If the operating steady state exists ($\frac{\rho_e}{H(0)} < 0$) and $H(0) < 0$, then $\rho_e > 0$, meaning that the shutdown steady state is linearly unstable.

2.3. Nonlinear stability conditions

In previous nonlinear stability studies, three important results, which will be used in Section 6, were derived: Welton's global asymptotic stability condition (Akcasu et al., 1971; Hetrick, 1971), Smets' boundedness condition (Smets, 1970) and Akcasu and Noble's finite escape time condition (Akcasu and Noble, 1966). Welton's condition is a sufficient condition for the global asymptotic stability of the operating steady state, given by $\mathrm{Re}\,[H(i\omega)] < 0$ for all ω (i.e. the real value of the feedback transfer function $H(s)$ is negative on the imaginary axis of the s plane). Global asymptotic stability means that the operating steady state is linearly stable and furthermore, any initial state in state space converges to it. Smets' condition is a sufficient condition for boundedness, given by $h(0) < 0$, meaning that the evolving states are always bounded in state space (do not tend to infinity). Akcasu and Noble's condition is a necessary condition for the existence of finite escape time solutions, given by $h(0) > 0$ and $H(0) < 0$, meaning that some solutions become unbounded after a finite duration. In both Smets' and Akcasu and Noble's conditions, $H(s)$ is assumed to have only poles with negative real value. In Smets (1972), it was shown that the Akcasu and Noble's finite escape time condition is also sufficient for a general G-L reactor model.

3. G-L reactor models in regular form

A G-L reactor model is a special case of the previously discussed point reactor model with linear reactivity feedback, having G delay groups and L temperature lumps (regions) with a linear temperature reactivity feedback for each temperature lump. Thus, the neutronics part of the model, composed of the point reactor equations with G delay groups, is given by Eqs. (1) and (2), while the reactivity equation in the G-L reactor model is a special case of the general finite-dimensional reactivity equation given in Eqs. (4) and (5). The reactivity equation can be decomposed into two parts: the thermal-hydraulics part, composed of the lumped-element temperature equations with L temperature lumps, and the algebraic reactivity equation part, containing a linear temperature reactivity feedback for each temperature lump.

In this section, the G-L reactor model will be introduced only in regular form. Previous studies have used G-L reactor models in other forms: temperature diagonal form, partially nondimensional form and degenerate form. These forms are presented and discussed in Appendix A.

For the G-L reactor model in regular form, the thermal-hydraulics part is given by linear lumped-element temperature equations with L temperature lumps:

$$C_{p,i}\frac{d}{dt}T_i(t) = \gamma_i P(t) + \sum_{j=1}^{L} K_{ij}T_j(t) + P_{e,i}\ , \tag{9}$$

where $C_{p,i}$ is the heat capacity of lump i, γ_i is the fraction of the reactor power deposited directly in the ith lump, K_{ij} is the thermal conductance[2] between lumps i and j and $P_{e,i}$ is the power into lump i from external regions (regions not included in the L temperature lumps).

By defining the following matrices and vectors: $[\mathbf{K}]_{ij} = K_{ij}$, $[\mathbf{C_p}]_{ij} = \delta_{ij}C_{p,i}$, $[\vec{T}]_i = T_i$, $[\vec{\gamma}]_i = \gamma_i$ and $[\vec{P}_e]_i = P_{e,i}$, Eq. (9) can be represented in vector form:

$$\frac{d}{dt}\vec{T}(t) = \boldsymbol{\sigma}\times\vec{T}(t) + \vec{\Gamma}_P P(t) + \vec{\Gamma}_e\ , \tag{10}$$

where $\boldsymbol{\sigma} = \mathbf{C_p^{-1}K}$, $\vec{\Gamma}_P = \mathbf{C_p^{-1}}\times\vec{\gamma}$ and $\vec{\Gamma}_e = \mathbf{C_p^{-1}}\times\vec{P}_e$.

The thermal conductance matrix $\mathbf{K}$ is assumed to be symmetric, $K_{ij} = K_{ji}$, weakly diagonally dominant, $|K_{ii}| \geq \sum_{j\neq i}|K_{ij}|$ for all $i = 1,\ldots,L$, and all the diagonal elements are negative $K_{ii} < 0$. This is indeed the case when the temperature lumps represent solid regions which are coupled by heat conductance, or when the lumps represent full fluid channels which are coupled to other solid lumps by heat convection. However, when a fluid channel is divided into axial segments (along the flow direction), then for the common situation of a flow with a high Péclet number, each axial segment is usually coupled only to the axial segment in the upstream direction (upwind scheme Patankar, 1980), hence $\mathbf{K}$ is not symmetric.

The algebraic reactivity equation is given by the following linear equation:

$$\rho(t) = \rho_e(t) + \vec{\alpha}\cdot\left(\vec{T}(t) - \vec{T}_0\right)\ , \tag{11}$$

where ρ_e is the external reactivity at zero power, $[\vec{\alpha}]_i = \alpha_i$ is the temperature reactivity coefficient of lump i and $\vec{T}_0$ is the temperature vector of the shutdown steady state, i.e. at zero power ($\vec{T}_0 = -\boldsymbol{\sigma}^{-1}\times\vec{\Gamma}_e$). It is assumed that the L temperature reactivity feedbacks are the only important power-reactivity feedback paths, at the relevant timescale of thermal-hydraulics (usually on the order of minutes or less).

For completeness, the full G-L reactor model in regular form is given by reiterating Eqs. (1), (2), (10) and (11):

$$\frac{d}{dt}P(t) = \frac{\rho(t)-\beta}{\Lambda}P(t) + \sum_{k=1}^{G}\lambda_k C_k(t)\ , \tag{12}$$

[2] Thermal conductance is the reciprocal of thermal resistance (has dimensions of power per unit temperature).

$$\frac{d}{dt}C_k(t) = \frac{\beta_k}{\Lambda}P(t) - \lambda_k C_k(t) \ ,$$
$$\frac{d}{dt}\vec{T}(t) = \boldsymbol{\sigma} \times \vec{T}(t) + \vec{\Gamma}_P P(t) + \vec{\Gamma}_e \ ,$$
$$\rho(t) = \rho_e(t) + \vec{\alpha} \cdot \left(\vec{T}(t) - \vec{T}_0\right) \ .$$

4. G-L reactor models: linear stability analysis

In this section, two results about general G-L reactor models are derived. First, it is shown that under plausible assumptions, the uncoupled thermal-hydraulics part of the G-L reactor model is linearly stable. Second, the codimension-1 bifurcations of the operating steady state, in which a loss of the linear stability of this state occurs, are characterized.

4.1. The uncoupled thermal-hydraulics part is linearly stable

As previously discussed, it is assumed that the thermal conductance matrix $\mathbf{K}$ is symmetric and weakly diagonally dominant. Moreover, it is assumed that $\mathbf{K}$ has at least one row r which is strictly diagonally dominant, $|K_{rr}| > \sum_{j\neq r}|K_{rj}|$, otherwise $\mathbf{K}$ is singular and the model is a degenerate G-L reactor model (discussed in Appendix A.3). Additionally, it can be assumed that the undirected graph associated with $\mathbf{K}$ is connected, which physically means that each pair of lumps is thermally coupled directly or via other lumps. Therefore, $\mathbf{K}$ is non-singular, i.e. has no zero eigenvalues (Horn and Johnson, 2012).

Because $\mathbf{K}$ is weakly diagonally dominant, according to Geršgorin disc theorem, all the real eigenvalues (real eigenvalues because $\mathbf{K}$ is symmetric) of $\mathbf{K}$ are non-positive, i.e. $\mathbf{K}$ is negative semi-definite (Horn and Johnson, 2012). Hence, all the real eigenvalues of the symmetric $\mathbf{K}$ are negative. As $\mathbf{K}$ is symmetric and $\mathbf{C_p^{-1}}$ is positive-definite, $\boldsymbol{\sigma} = \mathbf{C_p^{-1}K}$ is real-diagonalizable, $\boldsymbol{\sigma} = \mathbf{S}\boldsymbol{\sigma}'\mathbf{S}^{-1}$, where $\boldsymbol{\sigma}'$ is a real diagonal matrix having the same inertia as $\mathbf{K}$ (Horn and Johnson, 2012) (see Appendix A.1). Therefore, also $\boldsymbol{\sigma}$ has negative real eigenvalues.

Equivalently, it means that the feedback transfer function $H(s)$ has only negative real poles. In terms of stability, it means that the steady state of the uncoupled thermal-hydraulics part, i.e. Eq. (10) with a constant power $P(t) = P_1$, is linearly stable. This result constrains the thermal-hydraulic parameters ($\boldsymbol{\sigma}$) in G-L reactor models, and it shows that instabilities in these models can only come from the interaction between neutronics and thermal-hydraulics. This is in contrast to BWRs, where it is possible for an instability to be induced only via the uncoupled thermal-hydraulics, in the form of density wave oscillations (NEA, 1997).

4.2. Codimension-1 bifurcations of the operating steady state

The results in this section are applicable to the more general finite-dimensional point reactor models with linear reactivity feedback, for which all the poles of $H(s)$ have negative real value. Thus, in particular, the results are applicable to G-L reactor models, as it was just shown that all the poles of $H(s)$ are negative for G-L reactor models.

A bifurcation is an abrupt qualitative change in the state space dynamics, as the parameters of the model change (Guckenheimer and Holmes, 1983). The codimension of a bifurcation is the dimension of the parameter space minus the dimension of the bifurcation boundary. Thus, reaching a codimension-k bifurcation boundary requires, generically, to vary k parameters, called bifurcation parameters.

There are two generic types of codimension-1 local bifurcations of steady states: saddle–node bifurcation and Hopf bifurcation (Guckenheimer and Holmes, 1983). However, when special symmetries exists, there are more non-generic types of codimension-1 local bifurcations. It will now be shown that the static stability boundary, $H(0) = 0$, is a non-generic codimension-1 bifurcation of the operating steady state.

By writing the zero-power transfer function as $\bar{G}_0(s) = \frac{\bar{G}_0^N(s)}{\bar{G}_0^D(s)}$ and the feedback transfer function as $H(s) = \frac{H^N(s)}{H^D(s)}$ (where $\bar{G}_0^N(s)$, $\bar{G}_0^D(s)$, $H^N(s)$ and $H^D(s)$ are polynomials in s), the characteristic polynomial $p(s)$ of the operating steady state can be written by expanding the characteristic equation of the steady state, $1 + \bar{L}(s) = 0$, where $\bar{L}(s) = -P_1\bar{G}_0(s)H(s)$ is the open-loop transfer function:

$$p(s) = \bar{G}_0^D(s)H^D(s) - P_1\bar{G}_0^N(s)H^N(s) \ . \tag{13}$$

The constant term of the polynomial, a_n (assuming dimension n of the state space), is given by setting $s = 0$ in $p(s)$:

$$a_n = \bar{G}_0^D(0)H^D(0) - P_1\bar{G}_0^N(0)H^N(0) = -P_1\bar{G}_0^N(0)H^D(0)H(0) \ , \tag{14}$$

where the second equality is due to $\bar{G}_0^D(0) = 0$, because the zero-power transfer function has a pole at zero. $P_1 > 0$, $\bar{G}_0^N(0) > 0$ due to Eq. (7), and furthermore $H^D(0) > 0$, as it is assumed that all the poles of $H(s)$ have negative real value. It follows that $a_n = 0$ if and only if $H(0) = 0$, and $a_n > 0$ if and only if $H(0) < 0$. Generally, it can be shown that if $a_n = 0$ and $\frac{d}{d\lambda}a_n \neq 0$, then a single real root crosses the imaginary axis, where λ is the bifurcation parameter (Clarke and Jiang, 1993). Therefore, the static stability boundary is a bifurcation boundary in parameter space: when it is crossed (transversally), the operating steady state loses its linear stability, with a growing non-oscillatory mode.

Generically, when a single real root of the characteristic equation crosses the imaginary axis, it is a saddle–node bifurcation. However, the bifurcation at the static stability boundary $H(0) = 0$ is not a generic saddle–node bifurcation, but instead, will be called a pseudo transcritical bifurcation. While in generic saddle–node bifurcations two steady states collide and annihilate each other, in a transcritical bifurcation the two states collide and exchange their linear stability. Indeed, as the $H(0) = 0$ boundary is crossed, the shutdown and operating steady state exchange their linear stability (because if the operating steady state exists, then $\rho_e > 0$ and $H(0) < 0$, or $\rho_e < 0$ and $H(0) > 0$). However, the shutdown and operating steady states are not colliding in the usual sense, because for $H(0) < 0$ and $H(0) > 0$ there is always a finite distance in state space between the steady states, for arbitrarily small $|H(0)|$. At $H(0) = 0$, if the operating steady state exists, then $\rho_e = 0$ (due to $\rho_1 = \rho_e + P_1H(0) = 0$), meaning that there are infinitely many operating steady states with arbitrary $P_1 > 0$.

In Fig. 2, schematic bifurcation diagram of the pseudo transcritical bifurcation is shown. It should be noted that as the bifurcation variable $H(0)$ is varied, P_1 is constant and thus ρ_e also varies accordingly: $\rho_e = -P_1H(0)$ (due to $\rho_1 = \rho_e + P_1H(0) = 0$).

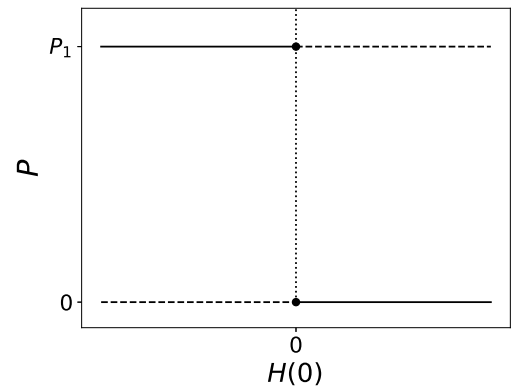


Fig. 2. Bifurcation diagram of the pseudo transcritical bifurcation, showing the stability of the shutdown and operating steady states, $P = 0$ and $P = P_1$, respectively. The horizontal axis is the bifurcation parameter $H(0)$ and the vertical axis is the power of the steady state. Linearly stable and unstable states are shown with solid and dashed lines, respectively. At $H(0) = 0$, there are infinitely many operating steady states with arbitrary power, shown in a dotted line.

If the bifurcation parameter is the effective parameter P_1,[3] then generically either $H(0) < 0$ or $H(0) > 0$. Changing P_1 will not change the value of $H(0)$, hence no real root can cross the imaginary axis. Meaning, if the single bifurcation parameter is the effective parameter P_1 and the static stability condition is satisfied, $H(0) < 0$, then the operating steady state can lose its linear stability only through a Hopf bifurcation. This result can also be derived utilizing standard linear control theory methods: root locus analysis or Nyquist plot analysis (Golnaraghi and Kuo, 2010).

5. G1-L2 reactor model: linear stability analysis

In this section, linear stability analysis of the G1-L2 reactor model is presented, i.e. a point reactor model with one delayed group and two temperature lumps. The G1-L2 reactor model is the most simplified reactor model that adequately capture the stability of nuclear reactors. Reactor models with $G = 0$, i.e. neglecting delayed neutrons, yield significantly different stability results compared to $G = 1$ (Schmidt and Hetrick, 1970). Additionally, it was found that including delayed neutrons in a reactor model does not always increase the domain of linear stability (in parameter space), contrary to what might be expected intuitively (Schmidt and Hetrick, 1970). Moreover, in most reactors there are at least two regions with non-negligible temperature reactivity coefficients, and thus a model with a single temperature lump, $L = 1$, cannot capture the full reactivity feedback mechanism. Indeed, reactor models with $L = 1$ have rather simple linear stability conditions (Hetrick, 1971).

In previous studies (Little and Schultz, 1957; Miida and Suda, 1961; Schmidt and Hetrick, 1970), the linear stability conditions were derived for the G1-L2 model only with approximations. In Section 5.1, complete linear stability conditions are derived for the G1-L2 reactor model in regular form, without any approximations. Furthermore, in Section 5.2, the derived linear stability region in the G1-L2 model will be classified based on qualitative properties.

It will be shown that linear instability in the G1-L2 reactor model is possible even when the two temperature reactivity coefficients are negative. Previous studies have also found that the intuitive requirement of only negative reactivity coefficients is not sufficient for linear stability of the operating steady state (Bethe, 1956; Schmidt and Hetrick, 1970). Indeed, in Schmidt and Hetrick (1970), examples are given for a G1-L2 reactor model which is linearly unstable when both temperature reactivity coefficients are negative but is linearly stable when one of them is positive.

5.1. Linear stability conditions

The equations for the G1-L2 reactor model in regular form are given by setting $G = 1$ and $L = 2$ in Eq. (12):

$$\begin{aligned}
&\frac{d}{dt}P(t) = \frac{\rho(t)-\beta}{\Lambda}P(t) + \lambda C(t)\ , \\
&\frac{d}{dt}C(t) = \frac{\beta}{\Lambda}P(t) - \lambda C(t)\ , \\
&\frac{d}{dt}T_1(t) = \sigma_{11}T_1(t) + \sigma_{12}T_2(t) + \Gamma_{P,1}P(t) + \Gamma_{e,1}\ , \\
&\frac{d}{dt}T_2(t) = \sigma_{21}T_2(t) + \sigma_{22}T_2(t) + \Gamma_{P,2}P(t) + \Gamma_{e,2}\ , \\
&\rho(t) = \rho_e(t) + \alpha_1\left(T_1(t) - T_{1,0}\right) + \alpha_2\left(T_2(t) - T_{2,0}\right)\ .
\end{aligned} \tag{15}$$

The elements of $\boldsymbol{\sigma}$ satisfy $\sigma_{11} = \frac{K_{11}}{C_{p,1}}$, $\sigma_{12} = \frac{K_{12}}{C_{p,1}}$, $\sigma_{21} = \frac{K_{21}}{C_{p,2}}$ and $\sigma_{22} = \frac{K_{22}}{C_{p,2}}$, where $K_{12} = K_{21}$, $-K_{11} \geq K_{12} > 0$, $-K_{22} \geq K_{21}$, and at least one of them satisfies a strict inequality, so that the reactor model is not degenerate (see Appendix A.3). Moreover, $\Gamma_{P,1} = \frac{\gamma_1}{C_{p,1}}$ and $\Gamma_{P,2} = \frac{\gamma_2}{C_{p,2}}$ are assumed to be nonzero. In a G1-L2 model, the temperatures will usually correspond to two of the three important regions in the reactor core: fuel, moderator or coolant (moderator is present only in thermal reactors).

As explained previously, the linear stability of the operating steady state of a G-L model can be studied using the characteristic equation $1+\bar{L}(s) = 0$, where $\bar{L}(s) = -\bar{G}_0(s)\bar{H}(s)$ is the open-loop transfer function and $\bar{H}(s) = P_1 H(s)$ is the normalized feedback transfer function. For the G1-L2 reactor model, the zero-power transfer function $\bar{G}_0(s)$ is given by setting $G = 1$ in Eq. (7):

$$\bar{G}_0(s) = \left[s\left(\Lambda + \frac{\beta}{s+\lambda}\right)\right]^{-1} = \frac{\lambda_p}{\beta}\frac{s+\lambda}{s\left(s+\lambda+\lambda_p\right)}\ , \tag{16}$$

where $\lambda_p = \beta\Lambda^{-1}$.

The normalized feedback transfer function $\bar{H}(s)$ is given by:

$$\bar{H}(s) = \frac{\left(\lambda_{\alpha,1}+\lambda_{\alpha,2}\right)s + \left(\sigma_{P,2}\lambda_{\alpha,1} + \sigma_{P,1}\lambda_{\alpha,2}\right)}{s^2 + t_\sigma s + d_\sigma^2}\ , \tag{17}$$

where $\lambda_{\alpha,i} = P_1\alpha_i\Gamma_{P,i}$, $\sigma_{P,1} = -\sigma_{11} + \frac{\Gamma_{P,1}}{\Gamma_{P,2}}\sigma_{21}$, $\sigma_{P,2} = -\sigma_{22} + \frac{\Gamma_{P,2}}{\Gamma_{P,1}}\sigma_{12}$, $t_\sigma = \mathrm{tr}\left[-\boldsymbol{\sigma}\right] = -\left(\sigma_{11}+\sigma_{22}\right)$ is the trace of $-\boldsymbol{\sigma}$, and $d_\sigma^2 = \mathrm{Det}\left[\boldsymbol{\sigma}\right] = \sigma_{11}\sigma_{22} - \sigma_{12}\sigma_{21}$ is the determinant of $\boldsymbol{\sigma}$ ($t_\sigma \geq 2d_\sigma$). As $K_{ii} < 0$, the parameters $\sigma_{P,1}$ and $\sigma_{P,2}$ are positive. The previously discussed static stability condition, $H(0) < 0$, is equivalent to $\sigma_{P,2}\lambda_{\alpha,1} + \sigma_{P,1}\lambda_{\alpha,2} < 0$.

The linear stability conditions are more intuitively understood in regular form and thus the linear stability analysis is carried in regular form (instead of, for example, temperature diagonal form, see Appendix A). The corresponding parameter space has 9 parameters, and they are chosen to be: λ_p, β, λ, t_σ, d_σ^2, $\sigma_{P,1}$, $\sigma_{P,2}$, $\lambda_{\alpha,1}$, $\lambda_{\alpha,2}$. The linear stability conditions define a non-trivial region in the parameter space. Hence, to simplify the analysis, two-dimensional cross-sections of the parameter space will be analyzed. The two parameters selected are $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, which are the only parameters that depend on the following three parameters: the operating power, P_1, and the two temperature reactivity coefficients, α_1 and α_2. These three parameters determine the strength of the reactivity feedback, and thus are highly important for the reactor stability. Note that as $\sigma_{P,i} > 0$, $\lambda_{\alpha,i} = P_1\alpha_i\Gamma_{P,i}$ have the same sign as the temperature reactivity coefficients α_i, respectively. In the parameter plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, the parameter $\lambda_{\alpha,1}$ is taken as the horizontal axis.

The characteristic equation $1+\bar{L}(s) = 0$, is given by the roots of the following fourth order polynomial:

$$p(s) = s^4 + a_1 s^3 + a_2 s^2 + a_3 s^1 + a_4\ , \tag{18}$$

where the expressions for the coefficients a_i are omitted for brevity. The Routh–Hurwitz criterion is necessary and sufficient for the characteristic polynomial to have roots with negative real value. For a fourth dimensional polynomial, in the Liénard–Chipart form (Gantmakher, 2000), it is given by the following four conditions: $a_1 > 0$, $a_2 > 0$, $a_4 > 0$ and $\Delta_3 = \Delta_2 a_3 - a_1^2 a_4 > 0$, where $\Delta_2 = a_1 a_2 - a_3$. The condition $a_1 > 0$ is always satisfied as it contains only positive terms. Hence, the three remaining non-trivial conditions are: $a_2, a_4, \Delta_3 > 0$. In the plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, both $a_2 > 0$ and $a_4 > 0$ specify a linear region, i.e. a region with a linear line boundary, whereas $\Delta_3 > 0$ specifies a hyperbolic region, a region for which the two branches of a hyperbola are its boundaries. The linear stability conditions are factored into terms with different non-positive powers of λ_p (so that applying the prompt jump approximation, $\lambda_p \to \infty$, is straightforward, see Appendix B). Note that in all derived expressions, there is a symmetry for changing the indices 1 and 2.

The linear region defined by $a_2 > 0$ is given by $\lambda_{\alpha,1} + \lambda_{\alpha,2} < \beta t_\sigma\left[1 + \left(\lambda + t_\sigma^{-1}d_\sigma^2\right)\lambda_p^{-1}\right]$, with a slope of $m_{a_2} = -1$. The linear region defined by $a_4 > 0$ is given by the static stability condition: $\sigma_{P,2}\lambda_{\alpha,1} + \sigma_{P,1}\lambda_{\alpha,2} < 0$, with a slope of $m_{a_4} = -\frac{\sigma_{P,2}}{\sigma_{P,1}}$. The $a_4 = 0$ line will be called in short the static line.

[3] The operating power, P_1, can be used as an effective parameter, which substitutes the external reactivity parameter, ρ_e, through $P_1 = -\frac{\rho_e}{H(0)}$.

The condition $\Delta_3 > 0$ can be written as a quadratic inequality, a conic section in $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$:

$$\Delta_3 = A_{11}\lambda_{\alpha,1}^2 + A_{12}\lambda_{\alpha,1}\lambda_{\alpha,2} + A_{22}\lambda_{\alpha,2}^2 + B_1\lambda_{\alpha,1} + B_2\lambda_{\alpha,2} + C > 0 \,, \tag{19}$$

where the coefficients A_{ij} and B_i are omitted for brevity. The discriminant of the conic section is given by $D = A_{12}^2 - 4A_{11}A_{22}$. Assuming $\sigma_{P,1} \neq \sigma_{P,2}$, it can be shown that $D > 0$ and therefore $\Delta_3 = 0$ is generically a hyperbola in $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$ (Rutter, 2000).

Hopf bifurcation occurs if $\Delta_{n-1} = 0$ and $\frac{d}{d\lambda}\Delta_{n-1} \neq 0$ (λ is the bifurcation parameter), when the rest of the Routh–Hurwitz conditions are satisfied (Kahoui and Weber, 2000). For the G1-L2 reactor model, it means that, in the plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, the intersection of the hyperbola $\Delta_3 = 0$ with the two other linear regions is the Hopf bifurcation boundary. When $\Delta_3 = 0$, generically $\frac{d}{d\lambda}\Delta_3 \neq 0$, as it represents intersection of a line and a hyperbola, if the bifurcation parameter is $\lambda_{\alpha,1}$ or $\lambda_{\alpha,2}$. The intersection of the static line, $a_4 = 0$, with the hyperbola and the linear region, $a_2 > 0$, is the pseudo transcritical bifurcation boundary, as explained in Section 4.2.

In the plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, the linear stability region, denoted here as the stable region, is the intersection of the two linear regions with the hyperbolic region. The boundary of the stable region is called the linear stability boundary. The part of the linear stability boundary that satisfies $a_4 = 0$ will be called the pseudo transcritical bifurcation boundary, or in short the static boundary, while the part that satisfies $\Delta_3 = 0$ will be called the Hopf bifurcation boundary.

5.2. Classification of the linear stability region

It can be shown that the static line, $a_4 = 0$, always intersects the hyperbola, $\Delta_3 = 0$, at two points, one satisfies $a_2 > 0$, the other $a_2 < 0$. Thus, there is a single intersection point of the static boundary and the Hopf boundary, which is given by:

$$\lambda_{\alpha,1,\mathrm{I}} = \begin{cases} \frac{\sigma_{P,1}}{\sigma_{P,1}-\sigma_{P,2}} \frac{\beta d_\sigma^2}{\lambda}\left(1 + \lambda\lambda_p^{-1}\right) & C'' < 0 \\ \frac{\sigma_{P,1}}{\sigma_{P,1}-\sigma_{P,2}} \beta t_\sigma \frac{1+(2\lambda+t_\sigma)\lambda_p^{-1}+(\lambda^2+\lambda t_\sigma+d_\sigma^2)\lambda_p^{-2}}{1+t_\sigma\lambda_p^{-1}} & C'' > 0 \end{cases} , \tag{20}$$

where $C'' = d_\sigma^2 - \lambda t_\sigma\left(1 + \lambda\lambda_p^{-1}\right)$ and $\lambda_{\alpha,2,\mathrm{I}}$ is given by changing the indices 1 and 2. For the first case, $C'' < 0$, it can be shown that at the intersection point there are two zero roots of the characteristic polynomial. For the second case, $C'' > 0$, it can be shown that at the intersection there are three roots with zero real value (two complex conjugate roots and one zero root).

Additionally, it can be shown that the intersections of the $a_2 = 0$ line with the static line or the hyperbola are in the unstable region, meaning that the stability boundary is composed only of the static boundary and the Hopf bifurcation boundary. This result is expected, because in general there are only two different possibilities for local codimension-1 bifurcations.

There are two different kinds of a general hyperbolic region, depending on the sign of $C' = C + \frac{A_{11}B_2^2 - A_{12}B_1B_2 + A_{22}B_1^2}{D}$. If $C' < 0$, then hyperbolic region is composed of two disjoint convex regions which are unbounded along the hyperbola transverse axis. On the other hand, if $C' > 0$, then the hyperbolic region is a single connected non-convex region which is unbounded along the hyperbola conjugate axis. For the G1-L2 hyperbolic region, C' is given by:

$$C' = \lambda\beta^2\left[d_\sigma^2 + \lambda\left(\lambda - t_\sigma\right)\right] \cdot \left[1 + 2\left(\lambda + t_\sigma\right)\lambda_p^{-1} + \left(\lambda + t_\sigma\right)^2\lambda_p^{-2}\right] , \tag{21}$$

such that $C' > 0$ is equivalent to $d_\sigma^2 + \lambda\left(\lambda - t_\sigma\right) > 0$. When $C' < 0$, the stable region is convex, as it is the intersection of (three) convex regions.

The slopes of the two hyperbola asymptotes can be calculated as follows: $m_1 = -\frac{\lambda+\sigma_{P,2}}{\lambda+\sigma_{P,1}}$ and $m_2 = -\frac{m_{22}}{m_{21}}$, where $m_{2i} = 1 + \left(t_\sigma - \sigma_{P,i}\right)\lambda_p^{-1}$. m_1 is always negative, while m_2 can be positive or negative.

The intersection of the bifurcation boundaries, hence also the Hopf bifurcation boundary, can be shown to always be below the hyperbola asymptote with slope m_1, and is below the asymptote with slope m_2 if $C' > 0$ and $m_{21} < 0$, or if $C' < 0$ and $m_{21} > 0$. As a consequence, the static boundary contains an unbounded upper part of the static line $a_4 = 0$ if $m_{21} < 0$ and $m_2 > 0$, or if $m_{21} > 0$ and $m_2 < m_{a_4} < 0$. It contains an unbounded lower part of the static line $a_4 = 0$ if $m_{21} > 0$ and $m_2 > 0$, or if $m_{21} > 0$ and $m_{a_4} < m_2 < 0$. The cases of $m_{21} < 0$ and $m_2 < 0$ are excluded, as it can be shown that $m_{21} < 0$ and $m_{22} < 0$ cannot occur simultaneously in the G1-L2 reactor model.

Therefore, the stable region can be classified according to three main parameters: m_2, m_{21} and C'. m_2 determines the asymptotic slope of the Hopf bifurcation boundary, having three qualitatively different cases: $m_2 > 0$, $m_{a_4} < m_2 < 0$, or $m_2 < m_{a_4} < 0$. m_{21} determines together with m_2 where the unbounded part of the static boundary resides: in the second or fourth quadrant. Finally, C' determines the convexity of the stable region: convex, $C' < 0$, or non-convex, $C' > 0$. The eight different possible types according to this classification are illustrated in Fig. 3.

To visually demonstrate how the stable region is formed, in Fig. 4 the stable region is shown, alongside the lines $a_2 = 0$ and $a_4 = 0$, and the hyperbola $\Delta_3 = 0$, for the region type of $C' < 0$, $m_2 > 0$ and $m_{21} < 0$. As was derived algebraically, it can be seen that the $a_2 = 0$ line is outside the linear stability boundary, and its role is to ensure that the other hyperbola branch is not included in the stability boundary.

6. Stability and bifurcation analysis of a G1-L2 reactor model: Generation IV HTR-type microreactor

Microreactors have electric power less than about 10 MW and are targeted to be used in remote or inaccessible environments. Several microreactors concepts are currently being designed (Testoni et al., 2021). The HTR type is the near-future focus of the (Generation IV) VHTR type, with coolant outlet temperatures of 700–950 °C.

In this section, we analyze the Holos-Quad reactor design, proposed by HolosGen, LLC and detailed in the 2019 report by Stauff et al. (2019). The Holos-Quad reactor design is still evolving: the current core design is called "Gen 2+" (Stauff et al., 2022). The Holos-Quad reactor is a Generation IV HTR-type microreactor, composed of four sub-critical power modules (SPMs), fits into a standard shipping container and has a nominal thermal power of 22 MW. It is a thermal reactor moderated by solid graphite and cooled by high temperature helium gas. Each fuel assembly (in the shape of a hexagonal prism) is made of a graphite matrix, with multiple cylindrical "holes" for fuel columns and coolant channels. The fuel column is also made of a graphite matrix, and is packed with about a million of TRISO (tri-structural isotropic) particles.

As described above, the G1-L2 reactor model is the simplest reactor model that adequately capture the stability of nuclear reactors. Indeed, in VHTR reactors, the coolant temperature reactivity feedback is generally negligible compared to the fuel and moderator temperature reactivity feedbacks, and thus $L = 2$ can be used in a simplified model of this reactor type.[4] Therefore, the Holos-Quad reactor is modeled using the G1-L2 simplified reactor model, where the two lumps (regions) are the fuel and moderator. To capture the macroscopic temperature profile, the fuel is lumped in the macroscopic level of the fuel columns and not the microscopic level of the TRISO particles.

This section demonstrates the application of our approach: utilizing both analytical and numerical methods to study the stability of a simplified model of a Generation IV reactor. In Section 6.1, the G1-L2 reactor model of the Holos-Quad microreactor is presented. In Section 6.2, the previously derived analytical linear stability results

[4] Using $L = 2$ is also applicable to the other three Generation IV types that G-L reactor models can model, which are fast reactors (GFR, LFR and SFR), in which the main temperature reactivity feedbacks are for the fuel and coolant.

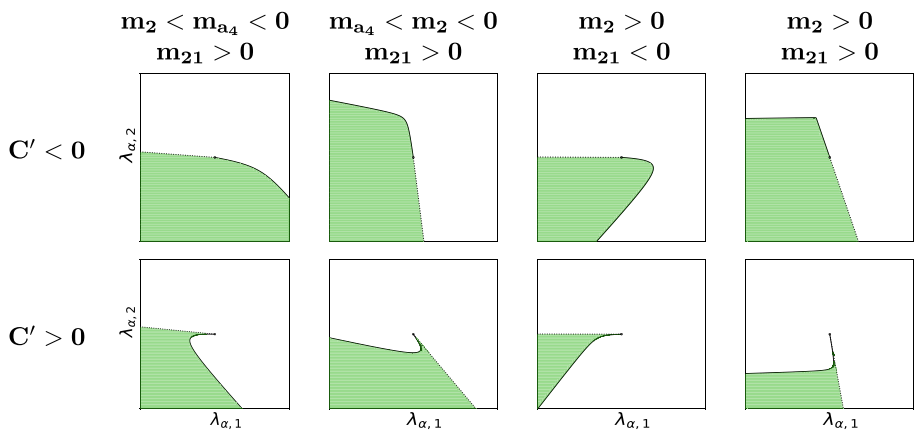


Fig. 3. Classification of the eight possible types of the linearly stable region in the G1-L2 reactor model, in the plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$. For each type, the Hopf bifurcation boundary appears in a black solid line, the static boundary (pseudo transcritical bifurcation boundary) appears in a black dotted line, and the stable region is colored in green. The intersection of the Hopf boundary and the static boundary appears as the black dot in the center.

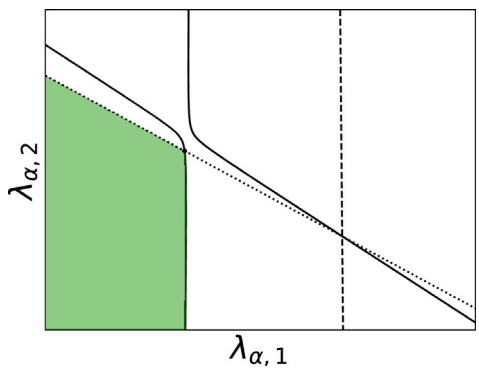


Fig. 4. The linearly stable region in the plane of $\lambda_{\alpha,1}$ and $\lambda_{\alpha,2}$, alongside the three linear stability curves, for the region type of $C' < 0$, $m_2 > 0$ and $m_{21} < 0$ (see Fig. 3). The hyberbola ($\Delta_3 = 0$) appears in a black solid line, the static line ($a_4 = 0$) appears in a black dotted line, the $a_2 = 0$ line appears in a dashed black line, and the stable region is colored in green.

are applied, along with nonlinear stability results from the literature (introduced in Section 2). Finally, in Section 6.3, local bifurcations of the G1-L2 reactor model are analyzed numerically using the numerical continuation method.

6.1. G1-L2 reactor model

The G1-L2 reactor model of the Holos-Quad reactor can be written in regular form, as:

$$\begin{aligned}
\frac{d}{dt}P(t) &= \frac{\rho(t)-\beta}{\Lambda}P(t) + \lambda C(t)\ , \\
\frac{d}{dt}C(t) &= \frac{\beta}{\Lambda}P(t) - \lambda C(t)\ , \\
C_{p,f}\frac{d}{dt}T_f(t) &= K_{fm}\left(T_m(t)-T_f(t)\right) + \gamma_f P(t)\ , \\
C_{p,m}\frac{d}{dt}T_m(t) &= K_{fm}\left(T_f(t)-T_m(t)\right) + K_{mc}\left(T_c - T_m(t)\right) + \gamma_m P(t)\ , \\
\rho(t) &= \rho_e(t) + \alpha_f\left(T_f(t)-T_{f,0}\right) + \alpha_m\left(T_m(t)-T_{m,0}\right)\ ,
\end{aligned} \tag{22}$$

where the subscripts are "f" for the fuel lump and "m" for the moderator lump. In this model, the coolant temperature T_c is regarded as a constant parameter, representing the average coolant temperature at the middle of the coolant channels. In more detailed models the coolant temperature will be represented by at least one dynamical variable.

In Table 1a, the estimated parameters of the G1-L2 reactor model of the Holos-Quad reactor are shown. The estimation of these parameters is mainly based on the design detailed in the report by Stauff et al. (2019), with some thermal-hydraulic parameters taken from NEA (2018). The single delay group decay constant λ was estimated using the approximation of $\lambda = \left(\frac{1}{\beta}\sum_k \frac{\beta_k}{\lambda_k}\right)^{-1}$. K_{fm} was estimated using an effective cylindrical geometry for the average fuel column and using one-dimensional steady state heat conductance formulas, and K_{mc} was estimated using the Dittus–Boelter equation for the convective heat transfer coefficient (Bergman and Lavine, 2017). It should be noted that the Dittus–Boelter correlation is not very accurate for strongly heated gas flows, such as the coolant flow in VHTR (Valentín et al., 2016), but it is used here out of simplicity. The value of T_c is not given as it is inconsequential to the dynamics of this model.

The G1-L2 reactor model equations in regular form can be partially nondimensionalized, i.e. all dimensions except the time dimension are removed, using the parameters that were introduced in Section 5:

$$\begin{aligned}
\frac{d}{dt}\delta\tilde{P}_1(t) &= \frac{\lambda_p}{\beta}\left[\delta\rho_{fb,f}(t)+\delta\rho_{fb,m}(t)\right]\left(1+\delta\tilde{P}_1(t)\right) - \lambda_p\delta\tilde{P}_1(t) + \lambda\delta\tilde{C}_1(t)\ , \\
\frac{d}{dt}\delta\tilde{C}_1(t) &= \lambda_p\delta\tilde{P}_1(t) - \lambda\delta\tilde{C}_1(t)\ , \\
\frac{d}{dt}\delta\rho_{fb,f}(t) &= \frac{d_\sigma^2 - \left(t_\sigma - \sigma_{P,m}\right)\sigma_{P,f}}{\sigma_{P,f}-\sigma_{P,m}}\delta\rho_{fb,f}(t) + \\
&\quad \frac{\lambda_{\alpha,f}}{\lambda_{\alpha,m}}\frac{d_\sigma^2 - \left(t_\sigma - \sigma_{P,m}\right)\sigma_{P,m}}{\sigma_{P,m}-\sigma_{P,f}}\delta\rho_{fb,m}(t) + \lambda_{\alpha,f}\delta\tilde{P}_1(t)\ , \\
\frac{d}{dt}\delta\rho_{fb,m}(t) &= \frac{\lambda_{\alpha,m}}{\lambda_{\alpha,f}}\frac{d_\sigma^2 - \left(t_\sigma - \sigma_{P,f}\right)\sigma_{P,f}}{\sigma_{P,f}-\sigma_{P,m}}\delta\rho_{fb,f}(t) + \\
&\quad \frac{d_\sigma^2 - \left(t_\sigma - \sigma_{P,f}\right)\sigma_{P,m}}{\sigma_{P,m}-\sigma_{P,f}}\delta\rho_{fb,m}(t) + \lambda_{\alpha,m}\delta\tilde{P}_1(t)\ .
\end{aligned} \tag{23}$$

As explained in the previous section, the most important parameters for stability are $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, as they are the only ones to contain the parameters P_1, α_f and α_m. The values for the partially nondimensional parameters, are shown in Table 1b. For $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, the nominal operating power $P_1 = 22$ MW was used.

6.2. Analytical stability analysis

The analytical linear stability results of Section 5 can be applied to the G1-L2 reactor model of the Holos-Quad reactor. Using these results, the linear stability region in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$ can be characterized. Using the parameter values of the Holos-Quad given in Table 1b, it is found that the linearly stable region type is given by $C' < 0$, $m_2 > 0$ and $m_{21} < 0$ (see Fig. 3).

The linearly stable region can be further characterized using the analytical nonlinear stability conditions that were introduced in Section 2:

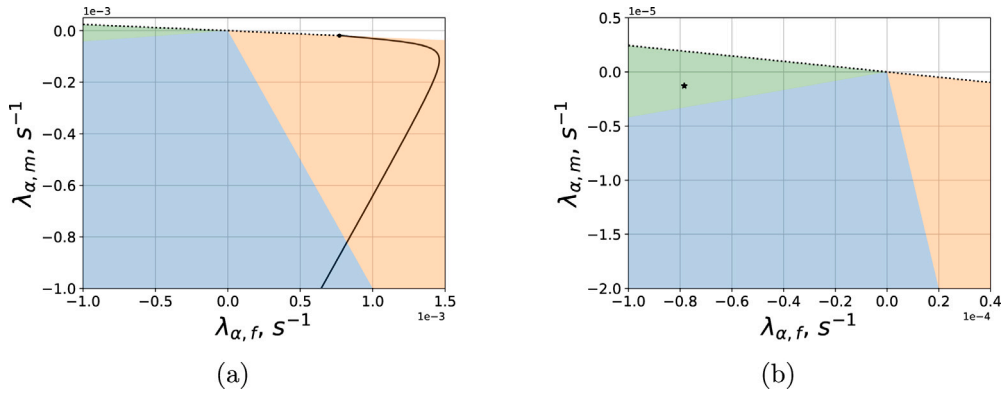


Fig. 5. Stability map for the G1-L2 model of the Holos-Quad reactor, in the parameter space of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$. Three regions are shown according to analytical nonlinear stability conditions: global asymptotic stability in green, boundedness in blue and finite escape time in orange. The Hopf bifurcation boundary appears in a black solid line and the static boundary (pseudo transcritical bifurcation boundary) appears in a black dotted line. The region in the left figure (a) is magnified in the right figure (b), around the nominal point of the Holos-Quad reactor, which is shown as a black star inside the global asymptotic stability region.

Table 1
The estimated parameters for a G1-L2 reactor model of the Holos-Quad reactor, dimensional and partially nondimensional.

(a) Dimensional parameters			(b) Partially nondimensional parameters		
Parameter	Value	Units	Parameter	Value	Units
Λ	$1.68 \cdot 10^{-3}$	s	λ_p	2.86	s^{-1}
β	$4.80 \cdot 10^{-3}$	–	β	$4.80 \cdot 10^{-3}$	–
λ	$8.86 \cdot 10^{-2}$	s^{-1}	λ	$8.86 \cdot 10^{-2}$	s^{-1}
$C_{p,f}$	$8.81 \cdot 10^{6}$	J/°C	t_σ	$4.07 \cdot 10^{-1}$	s^{-1}
$C_{p,m}$	$2.27 \cdot 10^{7}$	J/°C	d_σ	$1.16 \cdot 10^{-1}$	s^{-1}
K_{fm}	$2.25 \cdot 10^{6}$	W/°C	$\sigma_{P,f}$	6.38	s^{-1}
K_{mc}	$1.20 \cdot 10^{6}$	W/°C	$\sigma_{P,m}$	$1.56 \cdot 10^{-1}$	s^{-1}
γ_f	$9.60 \cdot 10^{-1}$	–	$\lambda_{\alpha,f}$	$-7.84 \cdot 10^{-5}$	s^{-1}
γ_m	$4.00 \cdot 10^{-2}$	–	$\lambda_{\alpha,m}$	$-1.29 \cdot 10^{-6}$	s^{-1}
α_f	$-3.27 \cdot 10^{-5}$	1/°C			
α_m	$-3.33 \cdot 10^{-5}$	1/°C			

Welton's global asymptotic stability condition, Smets' boundedness condition and Akcasu and Noble's finite escape time condition. Welton's global asymptotic stability condition, $\mathrm{Re}\,[H(i\omega)] < 0$ for all ω, is equivalent in the G1-L2 reactor model to the static stability condition and $\left(t_\sigma - \sigma_{P,m}\right)\lambda_{\alpha,f} + \left(t_\sigma - \sigma_{P,f}\right)\lambda_{\alpha,m} < 0$. Smets' boundedness condition, $h(t) < 0$, is given by $\lambda_{\alpha,f} + \lambda_{\alpha,m} < 0$ and $\left(2\sigma_{P,m} - t_\sigma + \sqrt{t_\sigma^2 - 4d_\sigma^2}\right)\lambda_{\alpha,f} + \left(2\sigma_{P,f} - t_\sigma + \sqrt{t_\sigma^2 - 4d_\sigma^2}\right)\lambda_{\alpha,m} < 0$. Finally, Akcasu and Noble's finite escape time condition, $h(0) > 0$, is expressed as $\lambda_{\alpha,f} + \lambda_{\alpha,m} > 0$, for the G1-L2 model (this condition is only valid when the static stability condition is satisfied). It is noted that Smets' boundedness condition and Akcasu and Noble's finite escape time condition are valid even when the operating steady state is linearly unstable.

In Fig. 5, the linearly stable region is shown, along with its partitioning to the three nonlinear stability conditions: global asymptotic stability in green (Welton's condition), boundedness in blue (Smets' condition) and finite escape time in orange (Akcasu and Noble's condition). There is an overlap between Welton's condition (green) and Smets' condition (blue), but it is shown only in green because global asymptotic stability also means boundedness. Moreover, the Hopf bifurcation boundary appears in a black solid line and the static boundary (pseudo transcritical bifurcation boundary) appears in a black dotted line.

Fig. 5(b) is a magnification of Fig. 5(a), in the vicinity the nominal point of the Holos-Quad reactor, $\lambda_{\alpha,m} = -7.9 \cdot 10^{-5}\ s^{-1}$ and $\lambda_{\alpha,f} = -1.3 \cdot 10^{-6}\ s^{-1}$ (see Table 1b), shown as a black star. The nominal point is inside the global asymptotic stability region, and in particular it is well inside the linearly stable region.

As Welton's condition is power-unconditional, and the nominal point satisfies it, it means that the Holos-Quad reactor in the G1-L2 model satisfies power-unconditional global asymptotic stability, and in particular power-unconditional linear stability. Geometrically, it means that in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, a ray from the origin that goes through the nominal point does not intersect the Hopf boundary, hence there is no critical power at which a loss of linear stability occurs.

6.3. Numerical bifurcation analysis

Although the nominal point of the Holos-Quad G1-L2 reactor model is globally asymptotically stable, it can be beneficial to study the stability and bifurcations in other parts of the parameter space. In this way, more knowledge can be gained about the nonlinear stability of G1-L2 reactor models. To this end, the Holos-Quad reactor is further analyzed using numerical continuation techniques, using the MatCont software (version 7.3) (Dhooge et al., 2008) for bifurcation analysis of dynamical systems.

Using numerical continuation, steady states or limit cycles can be continued (computed), as a single bifurcation parameter changes, and codimension-1 bifurcations can be detected. Additionally, the codimension-1 bifurcation points themselves can be continued, as two bifurcation parameters concurrently change, and codimension-2 bifurcations can be detected.

First, the Hopf codimension-1 bifurcation of the operating steady state was continued in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, yielding the Hopf bifurcation boundary that was derived analytically. During this continuation, the Hopf bifurcation type, supercritical or subcritical, is monitored. A generalized Hopf (GH) codimension-2 bifurcation was detected, which signifies the transition between supercritical and subcritical Hopf.

Second, the limit cycle emanating from the Hopf bifurcation was continued and limit point of cycles (LPC) codimension-1 bifurcation was detected. In the LPC bifurcation, also called saddle–node bifurcation of limit cycles, a stable limit cycle and an unstable (saddle) limit cycle collide and annihilate each other beyond a critical value of the bifurcation parameter. The LPC codimension-1 bifurcation was continued in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$. Two cusp point of cycles (CPC) codimension-2 bifurcations were detected, which signify the intersection point of two LPC curves in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$.

In Fig. 6, the Hopf and LPC codimension-1 bifurcation curves and the GH and CPC codimension-2 bifurcation points are shown in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, along with the static boundary and the three analytical nonlinear stability conditions from the previous section. The Hopf boundary appears in orange, its upper part is dashed, representing subcritical Hopf, and its lower part is solid, representing supercritical Hopf, with the GH point at the transition. The LPC curve is shown in blue, and is dashed for the case of a limit point of small-amplitude

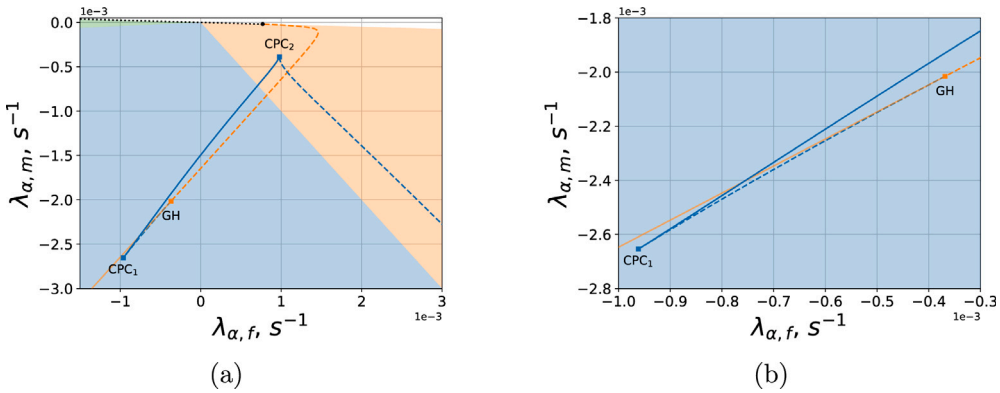


Fig. 6. Hopf and LPC bifurcation curves in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$. The Hopf curve (in orange) is dashed for subcritical Hopf and solid for supercritical Hopf. The LPC curve (in blue) is dashed when it represents a limit point of small-amplitude stable and larger-amplitude unstable limit cycles, and solid in the complementary case. The GH (orange) and CPCs (blue) codimension-2 bifurcation points are shown in squares. Also shown are the three regions due to the analytical nonlinear stability conditions, and the static boundary (pseudo transcritical bifurcation boundary), with the same conventions as in Fig. 5. The region in the left figure (a) is magnified in the right figure (b), around the region near the CPC_1 and GH points.

stable and larger-amplitude unstable limit cycles, and solid for the case of a limit point of small-amplitude unstable and larger-amplitude stable limit cycles. The LPC curve has three parts: the dashed part on the left, the solid part on the left and the dashed part on the right, which will be denoted by LPC_1, LPC_2 and LPC_3, respectively.

Fig. 6 can be better understood by taking sections of it with constant $\lambda_{\alpha,m}$, such that $\lambda_{\alpha,f}$ is the single bifurcation parameter. There are four different ranges of $\lambda_{\alpha,m}$ with qualitatively different dynamics: $\lambda_{\alpha,m}$ can be above the GH point and above or below the CPC_2 point, or it can be below the GH point and above or below the CPC_1 point. Fig. 7 presents bifurcation diagrams of the operating steady state and the limit cycles around it, for a constant $\lambda_{\alpha,m}$, with $\lambda_{\alpha,f}$ as the single bifurcation parameter. In these diagrams, the vertical axis is given by the maximum of the normalized power of the state, $\max(\delta\tilde{P}_1)$, where linearly stable and unstable states are shown with solid and dashed lines, respectively. Two cases out of the four mentioned cases are shown.

In Fig. 7(a), $\lambda_{\alpha,m} = -1.798 \cdot 10^{-3}\ s^{-1}$, such that $\lambda_{\alpha,m}$ is above the GH point and below the CPC_2 point, where a bistability of the operating steady state and a limit cycle occurs. As $\lambda_{\alpha,m}$ is above the GH point, the Hopf bifurcation type is subcritical, so an unstable limit cycle is born and exists to the left of the Hopf boundary in Fig. 6 (into the linearly stable region of the operating steady state). Because $\lambda_{\alpha,m}$ is below the CPC_2 point, this unstable limit cycle collides with a larger-amplitude stable limit cycle at LPC_2. Then, the stable limit cycle collides with a larger-amplitude unstable limit cycle at LPC_3. In this case, the loss of linear stability through the subcritical Hopf boundary results in a sudden transition to a large amplitude limit cycle. In the other case for which $\lambda_{\alpha,m}$ is above the GH point but above the CPC_2 point, the situation might be even more catastrophic, as no stable limit cycles were detected.

In Fig. 7(b), $\lambda_{\alpha,m} = -2.215 \cdot 10^{-3}\ s^{-1}$, such that $\lambda_{\alpha,m}$ is below the GH point and above the CPC_1 point, where a bistability of two limit cycles also occurs. As $\lambda_{\alpha,m}$ is below the GH point, the Hopf bifurcation type is supercritical, so a stable limit cycle is created and exists to the right of the Hopf boundary in Fig. 6 (into the linearly unstable region). Because $\lambda_{\alpha,m}$ is above the CPC_1 point, this stable limit cycle collides with a larger-amplitude unstable limit cycle at LPC_1. This larger-amplitude unstable limit cycle collides with a stable limit cycle with even larger amplitude at LPC_2, and finally collides with a larger-amplitude unstable limit cycle at LPC_3.

More generally, in the plane of $\lambda_{\alpha,f}$ and $\lambda_{\alpha,m}$, the region bounded by LPC_2, LPC_3 and the Hopf boundary is a bistability region of the operating steady state and a limit cycle, while the region bounded by LPC_1, LPC_2 and the Hopf boundary is a bistability region of two limit cycles.

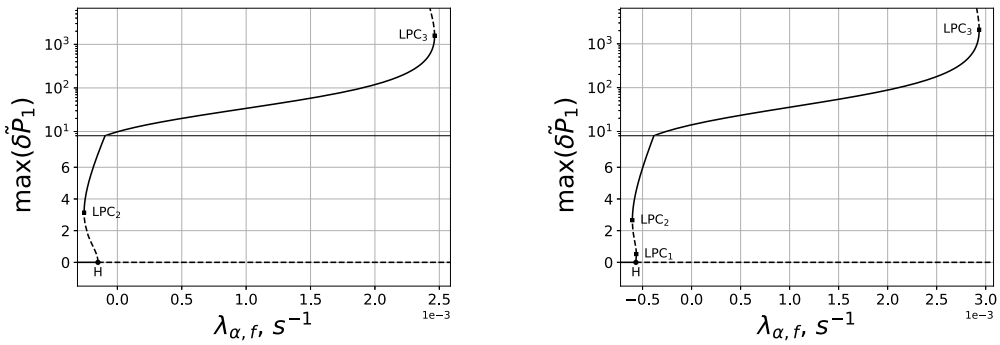
In all of the cases discussed above, the amplitude of the unstable limit cycle with the largest amplitude grows rapidly, and the continuation stops due to numerical convergence problems (it stops when the maximum nondimensional power $\delta\tilde{P}_1$ is on the order of $10^3 - 10^4$, see Fig. 7). There are two main options for this unstable limit cycle: it can continue indefinitely to the left (low values of $\lambda_{\alpha,f}$), or it can disappear in a bifurcation. The first option does not violate the boundedness and global asymptotic stability analytical conditions, as it is an unstable limit cycle. As for the second option, the unstable limit cycle can disappear in another bifurcation, which might be a LPC bifurcation, a bifurcation for which the amplitude of the limit cycle reaches infinity and thus disappears (Brøns and Sturis, 2001), or a homoclinic-to-hyperbolic-saddle bifurcation (Kuznetsov, 2004) (the saddle can be the shutdown steady state, or the operating steady state in the linearly unstable region).

As for the static line (pseudo transcritical bifurcation boundary), MatCont detects it as a branching point of two steady state branches, but it cannot perform branch switching, which is expected because this is not a regular transcritical bifurcation. The intersection of the Hopf boundary with the static boundary is detected as Bogdanov–Takens (BT) codimension-2 bifurcation by MatCont (Dhooge et al., 2008), due to the fact that at this point the operating steady state has two zero eigenvalues, as $C'' < 0$ for the G1-L2 model of the Holos-Quad (see Section 5.2). Nevertheless, this is not a generic BT bifurcation, as this point is not on a saddle–node bifurcation branch, but on a pseudo transcritical bifurcation branch.

In Pandey and Singh (2017), bifurcation analysis of a simplified reactor model of an advanced heavy water reactor was performed using MatCont. They analyzed a point reactor power model with $G = 1$, a fuel temperature lump and a void fraction dynamical variable, which can be shown to be equivalent to a constant power removal degenerate G1-L2 reactor model (see Appendix A.3). Indeed, they reached to bifurcation curves on a parametric plane which are qualitatively similar to those in Fig. 6. However, the fact that the detected BT point is not a generic BT point was not discussed, and the static stability boundary was not derived. Moreover, it can be shown that the CPC_2 point as well as the LPC_3 curve should exist, but they are missing in their parametric plane.

7. Conclusions

In this paper, the stability of G-L nuclear reactor models was analyzed. G-L reactor models can be used to perform stability analysis of simplified reactor models of four out of the six Generation IV reactor types (GFR, LFR, SFR and VHTR). This paper demonstrates the use of analytical methods to analyze the stability of Generation IV reactor models, in addition to the more usual numerical methods.



(a) $\lambda_{\alpha,m} = -1.798 \cdot 10^{-3}\mathrm{s}^{-1}$ (b) $\lambda_{\alpha,m} = -2.215 \cdot 10^{-3}\mathrm{s}^{-1}$

Fig. 7. Bifurcation diagrams of the maximum of the normalized power of the state, $\max(\delta\tilde{P}_1)$, against the single bifurcation parameter, $\lambda_{\alpha,f}$, showing bistability. Fig. 7(a) is for $\lambda_{\alpha,m} = -1.798 \cdot 10^{-3}$ s^{-1}, Fig. 7(b) is for $\lambda_{\alpha,m} = -2.215 \cdot 10^{-3}$ s^{-1}. Linearly stable and unstable states are shown with solid and dashed lines, respectively. The Hopf (H) bifurcation point is shown in a solid circle, and the LPC bifurcation points are shown in solid squares. Notice that the vertical axis is changing from linear scale to log scale.

In Section 4, two analytical linear stability results of general G-L reactor models were derived. First, it was shown that the thermal-hydraulics part of any G-L reactor model is linearly stable, and instability can appear only from the coupling to neutronics. Second, the well-known static stability boundary was characterized as a special codimension-1 bifurcation boundary, called a pseudo transcritical bifurcation.

In Section 5, linear stability of the G1-L2 reactor model was studied analytically, in regular form, without approximations. The linear stability region was classified into qualitatively different types, and its codimension-1 bifurcation boundaries were characterized.

Finally, in Section 6, stability and bifurcation analysis of a G1-L2 model of a Generation IV HTR-type microreactor was described. This analysis demonstrated the approach of studying a simplified reactor model of a Generation IV reactor from both the analytical and numerical perspectives, each having unique benefits. Applying the general analytical linear stability results of Section 5 and some general analytical nonlinear stability conditions from the literature, it was shown that the operating steady state, at the analyzed nominal point in parameter space, is not only linearly stable, but also globally asymptotically stable. Further applying numerical continuation, parts of the parameter space (relatively far from the nominal point) where stable and unstable limit cycles exist were detected, and a bistability of the operating steady state and a limit cycle, or of two limit cycles, was demonstrated.

To conclude, several future research directions are suggested. The nature of the pseudo transcritical bifurcation can be further analyzed, focusing on the (codimension-2) intersection of this bifurcation boundary (static stability boundary) and the Hopf bifurcation boundary. Another possible direction in this respect, is to generalize the analytical linear stability results derived in Section 5 for more detailed models, e.g. the G1-L3 model. The possible direction of deriving analytical conditions for the Hopf bifurcation type (supercritical or subcritical) of the Hopf boundary in the G1-L2 model (for example using the method of multiple time scales Nayfeh, 2011), is less attractive, because even for simpler models the derived analytical expressions are very complicated (Keener and Cohen, 1975; Wahi and Kumawat, 2011).

As another possible future direction, the bifurcation analysis via numerical continuation of the HTR-type microreactor presented in Section 6.3 may be elaborated and more bifurcations might be detected. In particular, this paper concentrated only on local bifurcations, while the analysis of global bifurcations, which involve the presence of homoclinic or heteroclinic connecting orbits (Kuznetsov, 2004), might enhance the understanding of the nonlinear dynamics in the G1-L2 model. Also, it can be interesting to explore other regions in the parameter space, and to study the sensitivity of the results to the model parameters that were assumed to be constant in Section 6.

Finally, the approach of stability and bifurcation analysis via both analytical and numerical methods, can be applied to simplified models of other Generation IV reactor types (GFR, LFR, SFR and VHTR), and of different reactor designs within each type. This can be done using the G1-L2 reactor model, as was done in this paper, or using more detailed models, such as the G1-L3 model and beyond.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. G-L reactor models: different forms

In this appendix, three different forms of G-L reactor models are presented and discussed. Previous studies have used these different forms, as they can simplify the analysis of G-L reactor models. Understanding the different forms of G-L reactor models might help interpreting previous stability studies and deriving new analytical stability results.

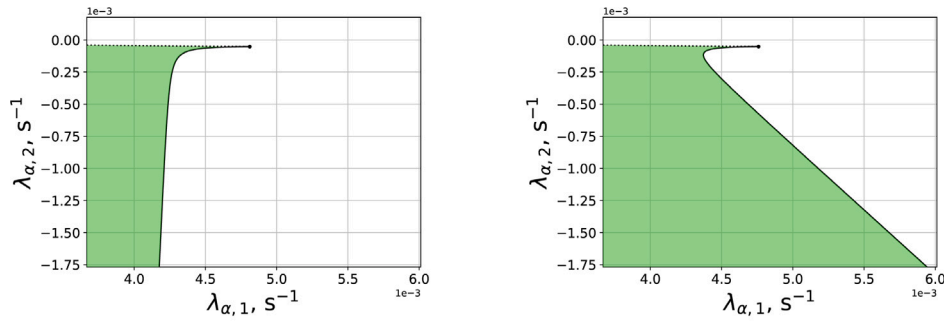
A.1. Temperature diagonal form

The lumped-element temperature equations, Eq. (10), can be cast into temperature diagonal form, in which the equations are diagonalized, i.e. the matrix $\boldsymbol{\sigma}$ is diagonalized over the reals by a real non-singular matrix $\mathbf{S}$:

$$\frac{d}{dt}\vec{T}'(t) = \boldsymbol{\sigma}' \times \vec{T}'(t) + \vec{\Gamma}'_P P(t) + \vec{\Gamma}'_e \ , \tag{A.1}$$

where $\boldsymbol{\sigma}' = \mathbf{S}^{-1}\boldsymbol{\sigma}\mathbf{S}$ is diagonal, $\vec{T}' = \mathbf{S}^{-1} \times \vec{T}$, $\vec{\Gamma}'_P = \mathbf{S}^{-1} \times \vec{\Gamma}_P$, and $\vec{\Gamma}'_e = \mathbf{S}^{-1} \times \vec{\Gamma}_e$. The existence of this real diagonalization transformation relies on the following theorem (Horn and Johnson, 2012): if $\mathbf{K}$ is symmetric and $\mathbf{C}_\mathbf{p}^{-1}$ is positive-definite, then $\boldsymbol{\sigma} = \mathbf{C}_\mathbf{p}^{-1}\mathbf{K}$ is diagonalizable, $\boldsymbol{\sigma} = \mathbf{S}\boldsymbol{\sigma}'\mathbf{S}^{-1}$, where $\boldsymbol{\sigma}'$ is a real diagonal matrix that has the same inertia as $\mathbf{K}$.

The G-L reactor model in temperature diagonal form was used in many studies, as it greatly simplifies the analytical stability analysis, e.g. Schmidt and Hetrick (1970) and Vreeke and Sandquist (1970). In many of these studies, the transformation was explicitly derived only for the case $L = 2$, but in Ergen et al. (1957) it was derived for general L, with the same assumptions of $\mathbf{K}$ symmetric and $\mathbf{C}_\mathbf{p}^{-1}$ positive-definite.



(a) Without PJA (b) With PJA

Fig. B.8. Example for a case where the PJA is not valid, with $\lambda_p > 10$ s^{-1}. The Hopf bifurcation boundary appears in a black solid line, the static boundary (pseudo transcritical bifurcation boundary) appears in a black dotted line, and the stable region is colored in green. The parameters are: $\lambda_p = 1.82 \cdot 10^1$ s^{-1}, $\beta = 5.48 \cdot 10^{-3}$, $\lambda = 9.81 \cdot 10^{-2}$ s^{-1}, $t_\sigma = 8.59 \cdot 10^{-1}$ s^{-1}, $d_\sigma = 3.51 \cdot 10^{-1}$ s^{-1}, $\sigma_{P,1} = 1.98 \cdot 10^1$ s^{-1}, $\sigma_{P,2} = 2.16 \cdot 10^{-1}$ s^{-1}.

A.2. Partially nondimensional form

Nondimensionalization can be applied to the G-L reactor model in order to reduce the number of parameters. Only a partial nondimensionalization will be applied, because there are many different time constants (parameters with dimension of time or inverse time).

Many different possible nondimensionalizations can be pursued, e.g. see the nondimensionalization used in Vreeke and Sandquist (1970). In this paper, a different nondimensional form is used, in which the dynamical variables are nondimensionalized as follows: $\delta\tilde{P}_1 = \frac{\delta P_1}{P_1}$, $\delta\tilde{C}_{k,1} = \frac{\delta C_{k,1}}{P_1}$, and $\delta\rho_{fb,i} = \alpha_i \delta T_{i,1}$, where the 0 subscript denotes the shutdown steady state, the 1 subscript denotes the operating steady state, and δ denotes deviation from steady state. $\delta\rho_{fb,i}$ represents the feedback reactivity contribution from the ith temperature lump, where it is assumed that all temperature lumps have nonzero temperature reactivity coefficients. The resulting nondimensional equations are:

$$\begin{aligned}
\frac{d}{dt}\delta\tilde{P}_1(t) &= \frac{\sum_{i=1}^{L} \delta\rho_{fb,i}(t) + \delta\rho_e(t)}{\Lambda}\left(1 + \delta\tilde{P}_1(t)\right) - \frac{\beta}{\Lambda}\delta\tilde{P}_1(t) \\
&\quad + \sum_{k=1}^{G} \lambda_k \delta\tilde{C}_{k,1}(t) \,, \qquad \text{(A.2)} \\
\frac{d}{dt}\delta\tilde{C}_{k,1}(t) &= \frac{\beta_k}{\Lambda}\delta\tilde{P}_1(t) - \lambda_k \delta\tilde{C}_{k,1}(t) \,, \\
\frac{d}{dt}\delta\rho_{fb,i}(t) &= \sum_{j=1}^{L} \frac{\alpha_i}{\alpha_j}\sigma_{ij}\delta\rho_{fb,j} + \lambda_{\alpha,i}\delta\tilde{P}_1(t) \,,
\end{aligned}$$

where $\lambda_{\alpha,i} = P_1 \alpha_i \Gamma_{P,i}$. These equations can be cast into a temperature diagonal form, further reducing the number of parameters.

A.3. Degenerate form

In a degenerate G-L reactor model, no row is strictly diagonally dominant, i.e. $-K_{ii} = \sum_{j\neq i} K_{ij}$ for $i = 1, \ldots, L$. Hence, the matrix $\mathbf{K}$, or equivalently $\boldsymbol{\sigma}$, is singular. In a non-degenerate model, for at least some lump i, $-K_{ii} > \sum_{j\neq i} K_{ij}$, and this lump allows power removal from the system, hence an operating steady state with nonzero power can be maintained. In the degenerate model, a power removal term should be added artificially, as a constant power removal term.

Generically, it is enough to add only one extra parameter, P_r, which represents a constant power removal term for lump i, so that $P_{e,j} = -\delta_{ij} P_r$ with $P_r > 0$. In this case, there is indeed a unique operating steady state, with power given by $P_1 = \frac{P_r}{\sum_{i=1}^{L} \gamma_i}$, but a shutdown steady state does not exist.

In a degenerate model, the feedback transfer function $H(s)$ has a simple pole at $s = 0$, thus the open-loop transfer function $\bar{L}(s)$ has a double pole (i.e. a pole with multiplicity 2) at $s = 0$. The static stability condition for a non-degenerate model, $H(0) < 0$, does not apply for the degenerate model, because it was derived by assuming only a simple pole at $s = 0$ (in fact, $H(0)$ is not defined in this case because of the pole at $s = 0$). A different static stability condition can be derived for the degenerate model, but it is more complicated: if $\bar{G}_0(s) = \frac{g(s)}{s}$ $(g(0) > 0)$ and $H(s) = \frac{h(s)}{s}$, with $g(s)$ and $h(s)$ analytic functions, then the static stability condition is given by $h(0) < 0$ and $\frac{d}{ds}\left[g(s)h(s)\right]_{s=0} > 0$.

Constant power removal degenerate G-L reactor models were studied in Ergen and Weinberg (1954), Little and Schultz (1957) and Shotkin (1964), only for the cases of one or two temperature lumps. The constant power removal model is useful as it yields less complicated analytical expressions for the linear stability conditions. However, a constant power removal model is usually an approximation to a non-constant, temperature-dependent, power removal. Therefore, in this paper, only non-degenerate reactor models were analyzed.

Appendix B. G1-L2 reactor model: prompt jump approximation

The prompt jump approximation (PJA) is the approximation of $\Lambda \to 0$ or $\lambda_p \to \infty$. In 1969–1971, several studies led by Hetrick, e.g. Schmidt and Hetrick (1970), derived linear stability conditions for a special case of the G1-L2 model in regular form, utilizing the PJA. In Hetrick (1971), the conditions were also derived for the G1-L2 model in temperature diagonal form, using the PJA. In Schmidt (1969), conditions for a special case of the G1-L2 in regular form, without using the PJA, were also derived, and it was noted that for very large values of $\lambda_p = \beta\Lambda^{-1}$, the PJA is a valid approximation. In Hetrick (1971), it was further argued that for realistic nuclear reactor parameter values of $\lambda_p > 10$ s^{-1}, the linear stability conditions using the PJA are "practically indistinguishable" from the exact G1-L2 conditions. However, as will be explained in this appendix, if the thermal-hydraulic time constants are fast enough, the PJA might not be a good approximation when Hetrick's condition $\lambda_p > 10$ s^{-1} is satisfied. Moreover, some thermal reactors have values of λ_p on the order of 1 s^{-1}.

Taking the PJA, the zero-power transfer function (with $G = 1$) is simplified to $\bar{G}_0(s) = \frac{s+\lambda}{s\beta}$, such that the $s = -(\lambda + \lambda_p)$ pole is neglected. A quick way to derive the linear stability conditions in the PJA, is to take the limit $\lambda_p \to \infty$ in all of the expressions derived in Section 5.2. The main properties of the stable region derived in Section 5.2 still hold in the PJA, but $m_2 = -1$ and thus the four types with $m_2 < 0$ are allowed, but not the four types with $m_2 > 0$ (see Fig. 3).

Approximate quantitative conditions for the validity of the PJA can be derived by examining the properties of the stable region, with and without the PJA. The static line and the hyperbolic region type ($C' \lessgtr 0$) are independent of λ_p, while the other hyperbola properties do depend on λ_p. The hyperbola can be characterized by the hyperbola center, the slopes of the hyperbola asymptotes - m_1 (which is independent on λ_p) and m_2, and the distance from the center to one of the vertices. Therefore, by requiring small ($\ll 1$) fractional deviations on these

parameters, the following approximate conditions for the validity of the PJA can be derived:

$$\left|\frac{\lambda^2 t_\sigma - \sigma_{P,i}\left(d_\sigma^2+\lambda^2\right)}{d_\sigma^2+\lambda\left(2\lambda-t_\sigma\right)+\sigma_{P,1}\left(\lambda-t_\sigma\right)}\right|, \left|t_\sigma-\sigma_{P,i}\right|, \lambda+t_\sigma \ll \lambda_p \ . \tag{B.1}$$

λ_p takes the typical values of 10^0 s^{-1} $\lesssim \lambda_p \lesssim$ 10^3 s^{-1} in thermal reactors, and 10^3 s^{-1} $\lesssim \lambda_p \lesssim$ 10^5 s^{-1} in fast reactors (Bell and Glasstone, 1970; Cervi et al., 2018). The value of λ is approximately 10^{-1} s^{-1} (for reactors based on U_{235}, Bell and Glasstone, 1970). The basic thermal-hydraulic time constants σ_{ij}, or t_σ and d_σ^2, are expected to be between 10^{-2} s^{-1} and 10^2 s^{-1}. One of the composite thermal-hydraulic time constants, $\sigma_{P,1}$ or $\sigma_{P,2}$, might be even larger by two orders of magnitude, as for example $\sigma_{P,1} = -\sigma_{11} + \frac{\Gamma_{P,1}}{\Gamma_{P,2}}\sigma_{21}$ include the term $\frac{\Gamma_{P,1}}{\Gamma_{P,2}} = \frac{\gamma_1}{\gamma_2}\frac{C_2}{C_1}$, where $\frac{\gamma_1}{\gamma_2}$ can be of order 10^2. All in all, it should be expected that the PJA will be more valid for fast reactors, assuming similar magnitude for the thermal-hydraulic time constants (t_σ, d_σ^2, $\sigma_{P,1}$ and $\sigma_{P,2}$).

The previously discussed PJA validity condition, $\lambda_p > 10$ s^{-1}, due to Hetrick (1971), will not hold for all possible reactor parameters in the ranges detailed above. First, because λ_p can be less than 10 s^{-1} in some thermal reactors (this is the case of the thermal microreactor detailed in Section 6). Second, even when $\lambda_p > 10$ s^{-1}, the PJA might not be a good approximation if some thermal-hydraulic time constants are fast enough. In Fig. B.8, an example is given for a case where the PJA is not valid, with $\lambda_p > 10$ s^{-1}. In this example, $\sigma_{P,1} - t_\sigma > \lambda_p$,[5] which means that $m_2 > 0$ without the PJA, as opposed to $m_2 < 0$ with the PJA.

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[5] In particular, the validity condition $\left|t_\sigma - \sigma_{P,1}\right| \ll \lambda_p$ is not satisfied.